🚀 Git Cheat Sheet

Beginner → Professional Quick Reference

⚙️ 1. Setup & Configuration

Action	Command
Check Git version	`git --version`
Set username	`git config --global user.name "Your Name"`
Set email	`git config --global user.email "you@email.com"`
Set default branch	`git config --global init.defaultBranch main`
Set default editor	`git config --global core.editor "code --wait"`
View all configs	`git config --list`

📁 2. Create or Clone a Repository

Action	Command
Create new repo	`git init`
Clone existing repo	`git clone <url>`
Clone into folder	`git clone <url> folder-name`

📊 3. Check Status & Differences

Action	Command
Show status	`git status`
Short status	`git status -s`

Action	Command
Show unstaged changes	`git diff`
Show staged changes	`git diff --staged`
Compare branches	`git diff branch1..branch2`

➕ 4. Stage & Commit

Action	Command
Stage one file	`git add file.txt`
Stage all files	`git add .`
Stage by patch	`git add -p`
Commit	`git commit -m "message"`
Stage + commit tracked files	`git commit -am "message"`
Amend last commit	`git commit --amend`

🌿 5. Branching

Action	Command
List branches	`git branch`
List all (incl. remote)	`git branch -a`
Create branch	`git branch name`
Switch branch	`git switch name`
Create & switch	`git switch -c name`
Rename branch	`git branch -m new-name`
Delete branch	`git branch -d name`
Force delete branch	`git branch -D name`

🔄 6. Merging & Rebasing

Action	Command
Merge branch into current	`git merge branch-name`
Rebase current onto branch	`git rebase branch-name`
Abort merge	`git merge --abort`
Abort rebase	`git rebase --abort`
Continue rebase	`git rebase --continue`

🌐 7. Remote Repositories

Action	Command
View remotes	`git remote -v`
Add remote	`git remote add origin <url>`
Remove remote	`git remote remove origin`
Rename remote	`git remote rename old new`
Fetch updates	`git fetch`
Pull updates	`git pull`
Push branch	`git push -u origin branch`
Push all tags	`git push --tags`

⏪ 8. Undoing Changes

Action	Command
Discard file changes	`git restore file.txt`
Unstage a file	`git restore --staged file.txt`

Action	Command
Undo last commit (keep changes)	`git reset --soft HEAD~1`
Undo last commit (unstage)	`git reset HEAD~1`
Undo last commit (delete changes)	`git reset --hard HEAD~1`
Revert a pushed commit	`git revert <commit-id>`

> ⚠️ Use `--hard` carefully — it deletes work permanently.

📜 9. History & Logs

Action	Command
Full log	`git log`
Compact log	`git log --oneline`
Graph view	`git log --oneline --graph --decorate --all`
Show one commit	`git show <commit-id>`
Who changed a line	`git blame file.txt`
Search commits	`git log --grep="keyword"`

📦 10. Stashing Work

Action	Command
Stash changes	`git stash`
Stash with message	`git stash push -m "message"`
List stashes	`git stash list`
Apply latest stash	`git stash apply`
Apply + remove stash	`git stash pop`
Drop a stash	`git stash drop stash@{0}`

Action	Command
Clear all stashes	`git stash clear`

🏷️ 11. Tags & Releases

Action	Command
Create tag	`git tag v1.0.0`
Annotated tag	`git tag -a v1.0.0 -m "Release"`
Push tag	`git push origin v1.0.0`
Push all tags	`git push --tags`
Delete local tag	`git tag -d v1.0.0`
Delete remote tag	`git push origin --delete v1.0.0`

🍒 12. Cherry-Pick & Reflog

Action	Command
Apply specific commit	`git cherry-pick <commit-id>`
View reflog (history of HEAD)	`git reflog`
Recover lost commit	`git checkout <commit-id>`

🧠 13. Useful Aliases (Optional)

```
git config --global alias.st status
git config --global alias.co checkout
git config --global alias.br branch
git config --global alias.cm commit
git config --global alias.lg "log --oneline --graph --decorate --all"
```